

Q. No: 1(a). Answer the Question No: 1 (a)

```
#include <stdio.h>
#include <math.h>
int main () {
    float n, m=5;
    scanf ("%f", &n);
    float p = (int n - 1*(int) m) / sqrt(6);
    printf ("%f", p);
    return 0;
}
```

Q. No: 1(b). Answer the Question No: 1 (b)

Answer to the Q.No: 1(b)

- ① Num1 → valid
- ② 1-100 → invalid → Number at first
- ③ \$ value → invalid → Special character
- ④ "myvalue" → invalid → " "
- ⑤ num6 → valid
- ⑥ first_name → valid
- ⑦ last_name → invalid → Special character
- ⑧ f1X → valid.

Answer to the Q.No: 1(c)

int a = 10.0 / 3 * 10	a = 33
float b = (int) 23.0 % 11	b = 1
int c = (10 > 9) 21 <= 19 * 5	c = 0
float d = 7/2	d = 3.0

Answer to the Q. NO: 2(a)

b=4 : output:

Begin
CSE
End

b=10 output:

Begin
VIU
End

b=5 output:

Begin
VIU
End

b=12 output:

Begin
VIU
End

Answer to the Question No: 2(b)

#include <stdio.h>

char c;

int exp, family;

float income;

getchar();

scanf("%c", &c);

scanf("%d", &exp);

scanf("%d", &family);

scanf("%f", &income);

int eg = 0;

if ((exp >= 12 && family > 5) || income > 1000.50)

 eligib eg = 1;

else if ((c == 'Y' || c == 'y') && family > 8 && income < 1100.78)

 eg = 1;

else if (c == 'X' && family > 6)

 eg = 1;

if (eg) { printf("Eligible for Bonus");

else { printf("Not eligible for Bonus");

Answer to Question No: 3 (a)

i	Sum	a	Sign
ganbadge	0	3	1
1	3	7	-1
2	-4	13	1
3	9	17	-1
4	-8	23	1
5	15	27	-1
6			

Answer to the Q. No: 3(b)

```
#include <stdio.h>

int main () {
    int n, i, sum = 0;
    scanf ("%d", &n);
    for (i = 1; i <= n/2; i++) {
        if (n % i == 0)
            sum += i;
    }
    if (sum < n) { printf ("Deficient Number\n"); }
    else { printf ("Not Deficient\n"); }
    return 0; }
```

Answer to the Q. No: 4(a)

i	f[i]
2	{1, 1, 2, 0, 0, 0}
3	{1, 1, 2, 3, 0, 0}
4	{1, 1, 2, 3, 5, 0}
5	{1, 1, 2, 3, 5, 8}

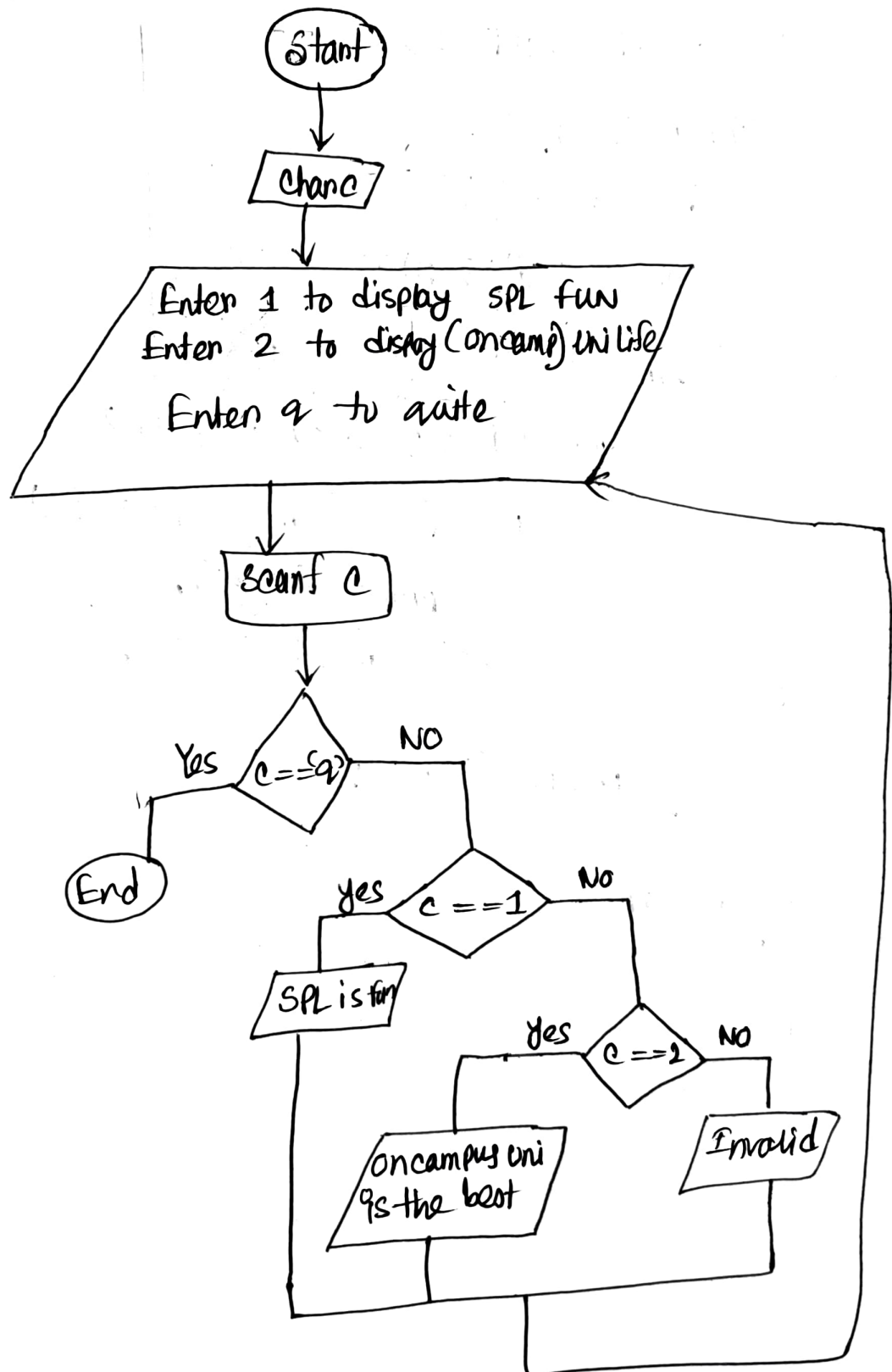
Answer to the Question No: 4(b)

```
#include <stdio.h>

int main() {
    int n, i;
    scanf("%d", &n);
    int arr[n];
    for (i=0; i<n; i++) {
        scanf("%d", &arr[i]);
    }
    int max = arr[0], min = arr[0];
    int maxIndex = 0, minIndex = 0;
    for (i=1; i<n; i++) {
        if (arr[i] > max) {
            max = arr[i];
            maxIndex = i;
        }
        if (arr[i] < min) {
            min = arr[i];
            minIndex = i;
        }
    }
}
```

printf("%d", i, d, max, maxIndex);
printf("%d", i, d, min, minIndex);
return 0;

Answer to the Question NO: 5(a)



Answer to the Question NO: 5(b)

```
#include <stdio.h>

int main () {
    int n, i, j;
    scanf ("%d", &n);
    for (i=0; i<n; i++) {
        for (j=0; j<n; j++) {
            if (j==i || j==n-i-1) {
                printf ("*");
            }
            else printf (" ");
        }
        printf ("\n");
    }
    return 0;
}
```


Answer to the Question No: 1(a)

```
#include <stdio.h>

int main() {
    int num1 = 5;
    float num2;
    char chn = 'a';
    scanf ("%f", &num2);
    num1 = (int) num2 % chn;
    printf ("Result is = %d", num1);
    return 0;
}
```

Answer to the Q. NO: 1(B)

longest_val → valid
 smallest-one → invalid → Special character.
 while → invalid → keyword.
 2ndNum → invalid → 1st number.
 ! New → ! not allowed → invalid.
 avg marks → invalid → space in the
 val 9 → valid

Answer to the Question No: 2(c)

float a = 5 * (5/2)	a = 10.0
int b = 5 * (5/2)	b = 10
float c = 5 * (5.0/2)	c = 12.5
int d = 5 * (5.0/2)	d = 12

Answer to the Question No: 2(a)

num=1 output: 12 5 12

num=3 output:
0 0 0

Answer to the Q. NO: 2(b)

i	Sum	a	b	2	y
1	1	8	7	3	2
2	9	27	19	6	3
3	36	64	27	10	4
4	100	125	61	15	5
5	225	216	91	21	6
6					

Answer to the Q. NO: 2(a)

```
void main () {
```

```
    int n=3, i, j, sum=0;
```

```
    i=0;
```

```
    do { j=0;
```

```
        do {
```

```
            if (i==j) sum += i+j;
```

```
            else if (i>j) sum += i+n;
```

```
            else sum += n-j;
```

```
            j++; } while (j<=1)
```

```
            i++; } while (i<n); }
```


Answer to the Q. NO: 8(b)

```
#include <stdio.h>
```

```
int main () {
```

```
    int num, count = 0;
```

```
    float sum = 0, avg;
```

```
    while (1) {
```

```
        printf ("Enter a positive Number (0 to stop): ");
```

```
        scanf ("%d", &num);
```

```
        if (num == 0)
```

```
            break;
```

```
        if (num > 0) {
```

```
            sum += num;
```

```
            count ++; }
```

```
    }
```

```
    if (count > 0) {
```

```
        avg = sum / count;
```

```
        printf ("Avg = %.2f \n", avg);
```

```
    }
```

```
    else {
```

```
        printf ("No positive Number \n");
```

```
    }
```

```
    return 0;
```

```
}
```

Answer to the Question No: 4(a)

n	i	f[6]
2	0	{ 3, 0, 0, 0, 0, 0 }
3	1	{ 3, 8, 0, 0, 0, 0 }
3	2	{ 3, 8, 5, 0, 0, 0 }
3	3	{ 3, 8, 5, 12, 0, 0 }
3	4	{ 3, 8, 5, 12, 7, 0 }
3	5	{ 3, 8, 5, 12, 7, 16 }

Answer to the Question No: 4(b)

```
#include <stdio.h>
```

```
int main () {
```

```
    int arr[100], n, i, sum=0, count=0;
```

```
    float avg;
```

```
    printf ("Enter the value of n: ");
```

```
    scanf ("%d", &n);
```

```
    printf ("Enter %d integers", n);
```

```
    for (i=0; i<n; i++) { scanf ("%d", &arr[i]);
                           printf ("%d Index \t value \n", i, arr[i]);
                           printf ("-----\n");
    }
```

```
    for (i=0; i<n; i++)
        printf ("%d\t %d\n", i, arr[i]);
```

```
    for (i=1; i<n; i+=2) {
        sum += arr[i];
        count++;
    }
```

```
if (count > 0) {
```

```
    avg = (float) sum / count;
```

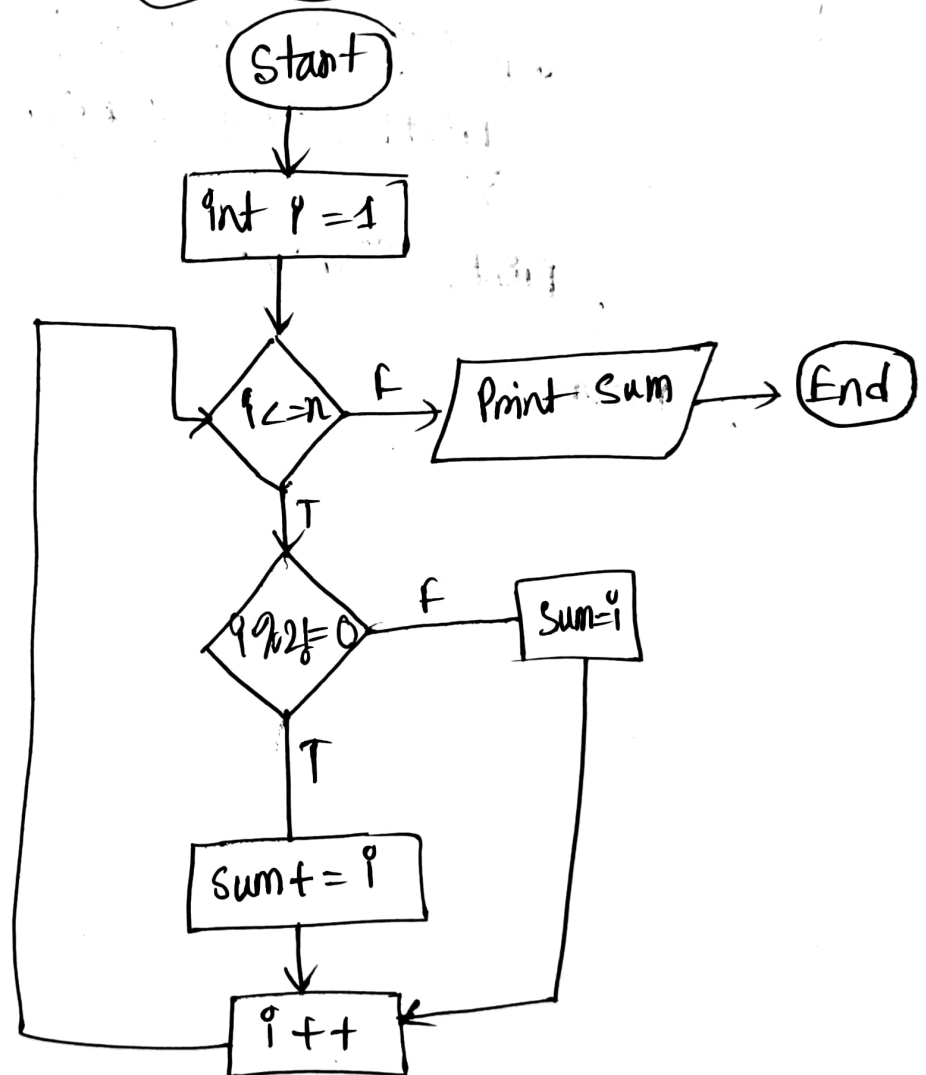
```
    printf ("Avg num at odd index : %2f", avg);
```

```
else
```

```
    printf ("No Element exist at odd to cal avg\n");
```

```
}
```

Answer to the Q. No: 5(a)



Answer to the Q.No: 5(b).

```
#include <stdio.h>
```

```
int main () {
```

```
    int i, j, n;
```

```
    scanf ("%d", &n);
```

```
    for (i=n; i>=1; i--) {
```

```
        for (j=i; j>=1; j--) {
```

```
            printf ("%d", j*2);
```

```
        }
```

```
        printf ("\n");
```

```
    }
```

```
    return 0;
```

```
}
```